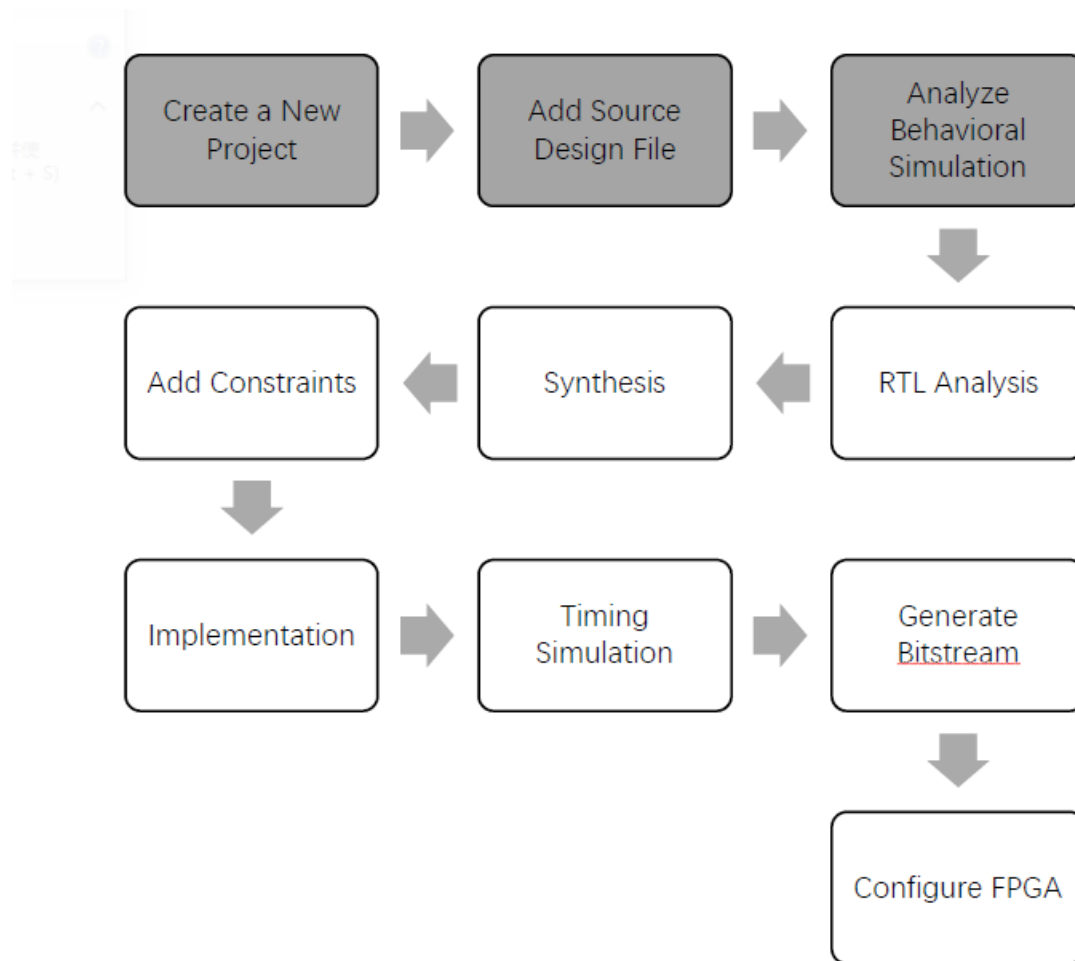


Vivado Tutorial

Introduction

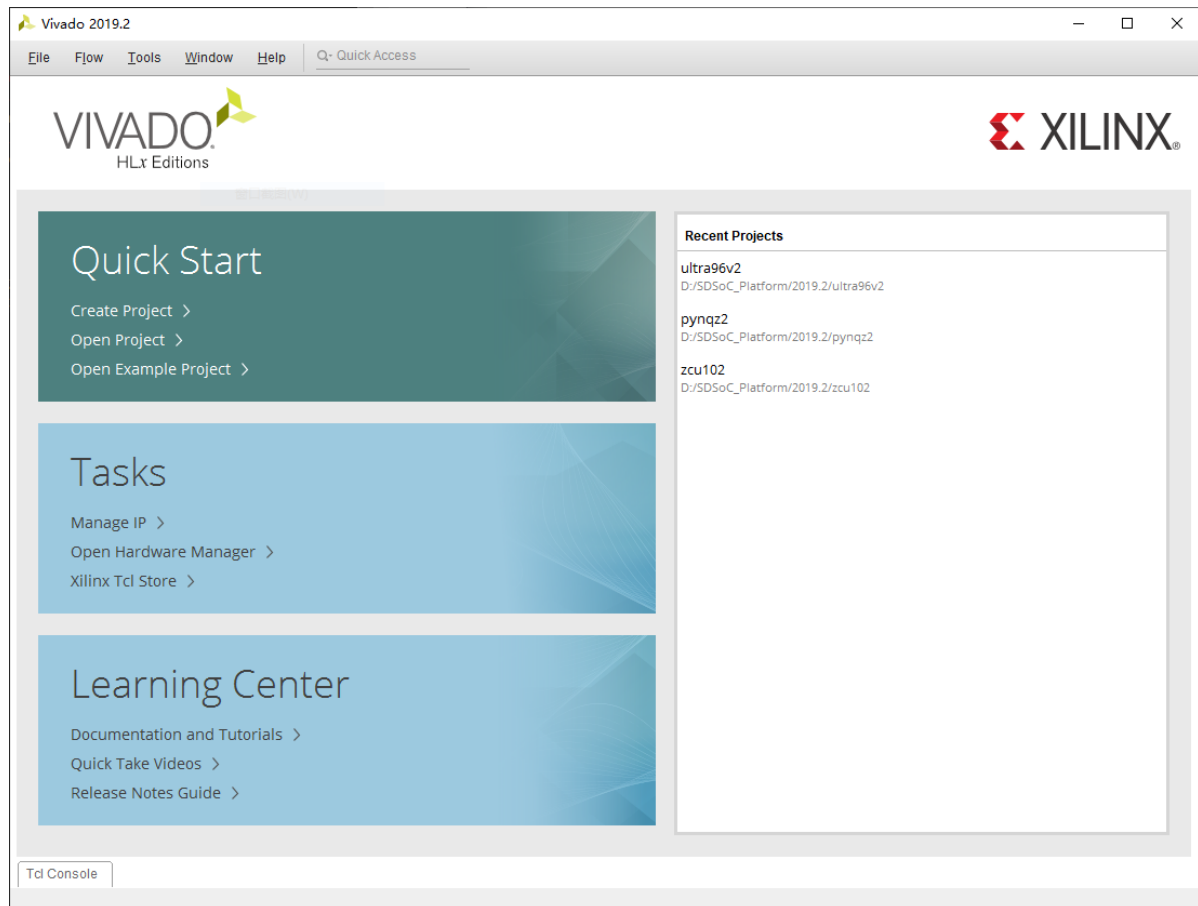
In this lab, we will guide you to create a simple digital design using Xilinx Vivado software. A typical digital design flow consists of creating new projects, adding source design file, analyzing behavioral simulation, RTL analysis, synthesis, adding constraints, implementation, timing simulation, generating bitstream and configuring FPGA. You will go through the typical design flow targeting the Zynq-7000 SoC. The typical design flow is shown below. We focus on the first three steps in this tutorial.



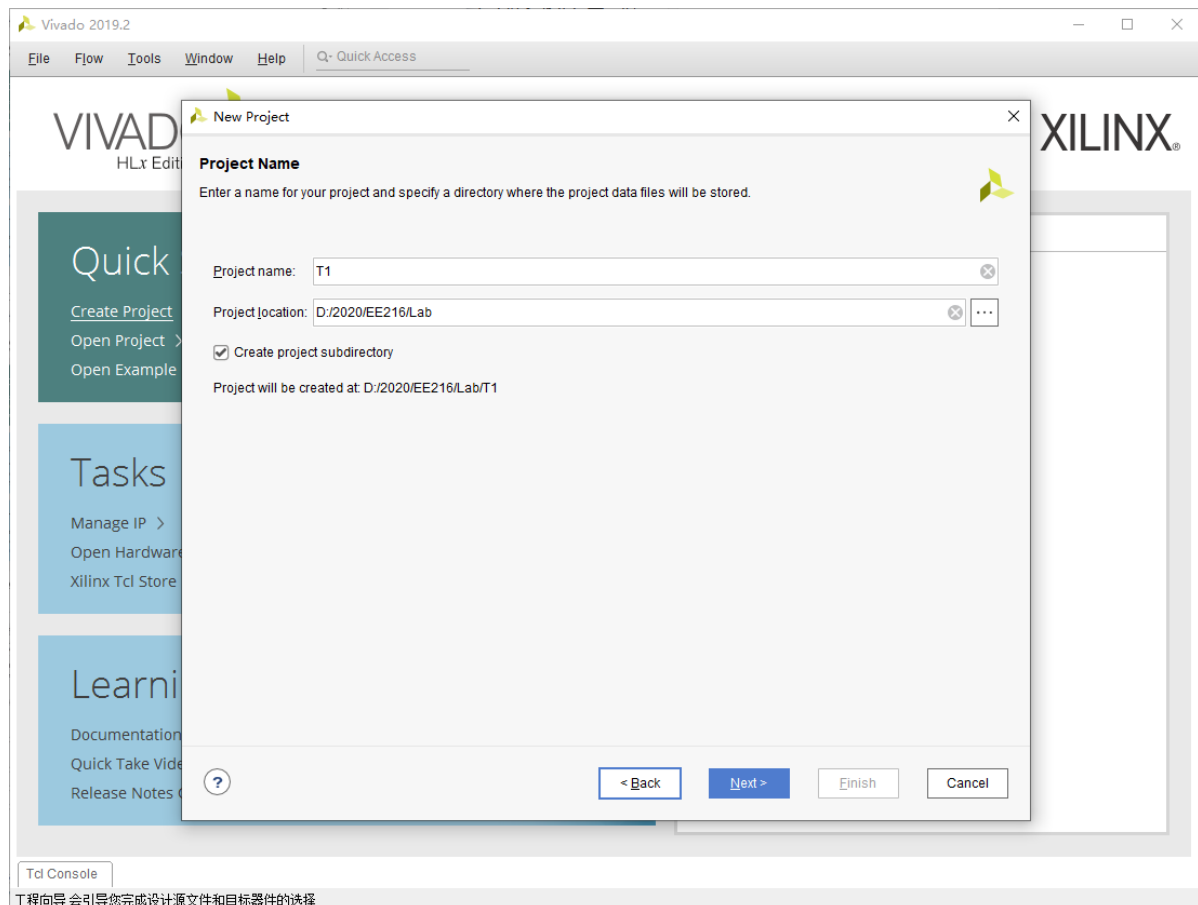
General Flow

Create A New Project

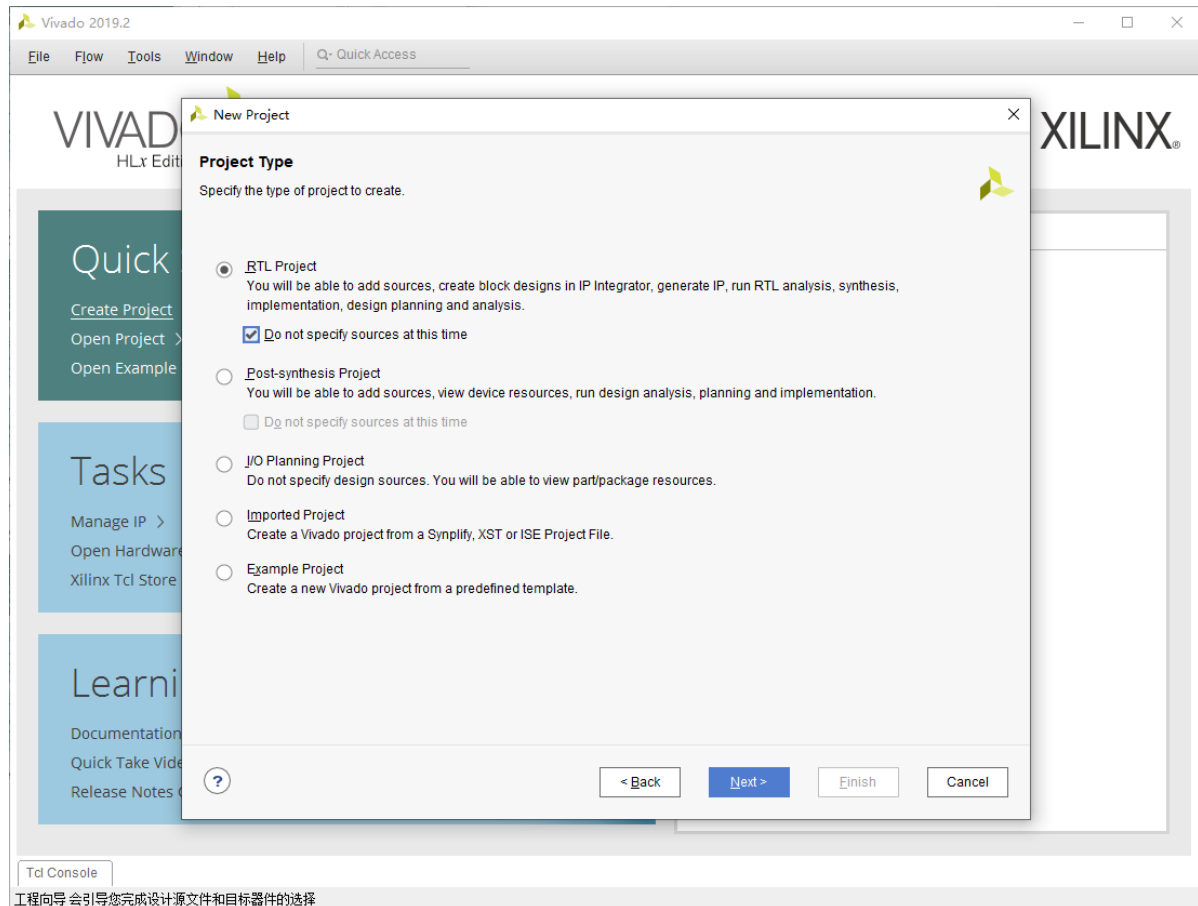
1. **Open Vivado** by selecting Start > All Programs > Xilinx Design Tools > Vivado 2020.2 > Vivado 2020.2
2. Click **Create New Project** to start the wizard. You will see Create A New Vivado Project dialog box. Click Next.



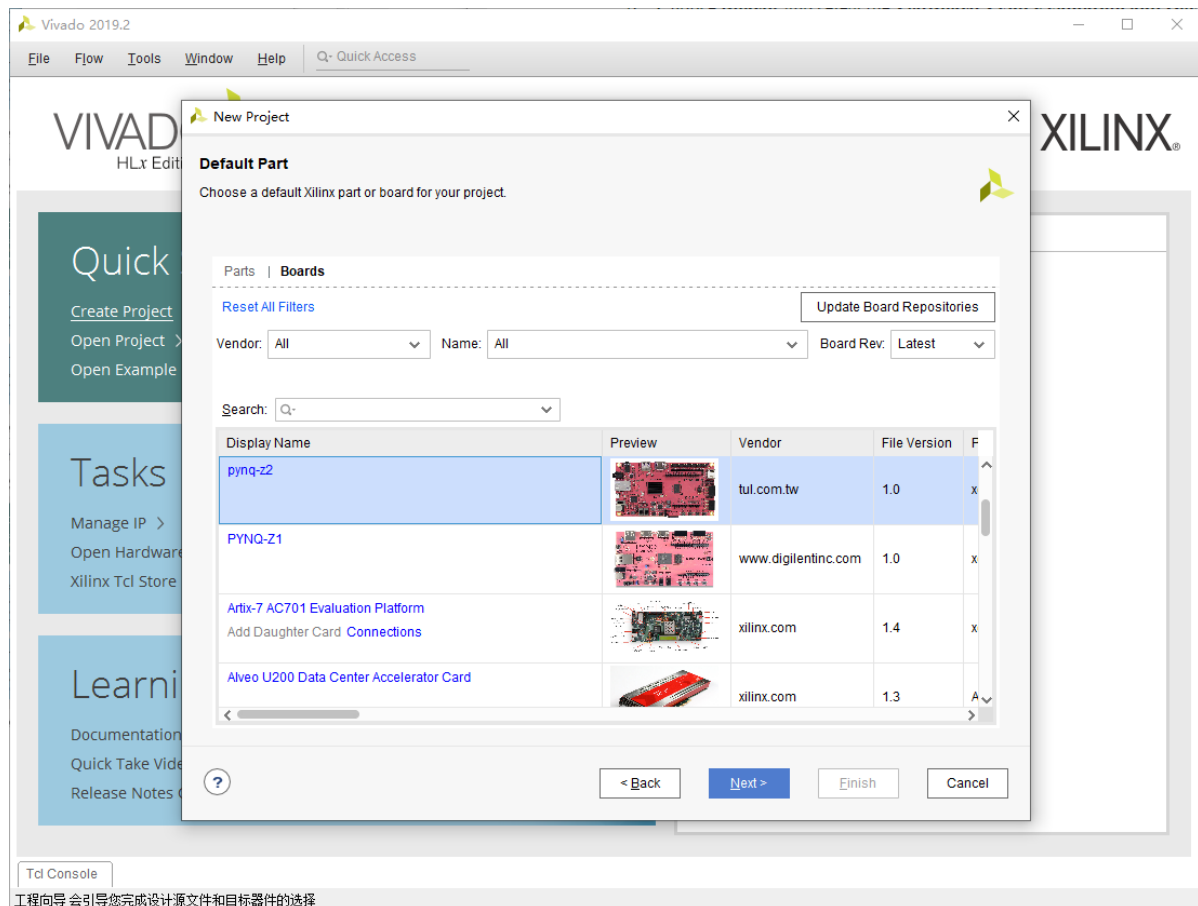
3. Fill the **Project location** field.
4. Enter tutorial in the **Project name** field. Make sure that the **Create Project Subdirectory** box is checked. Click Next.



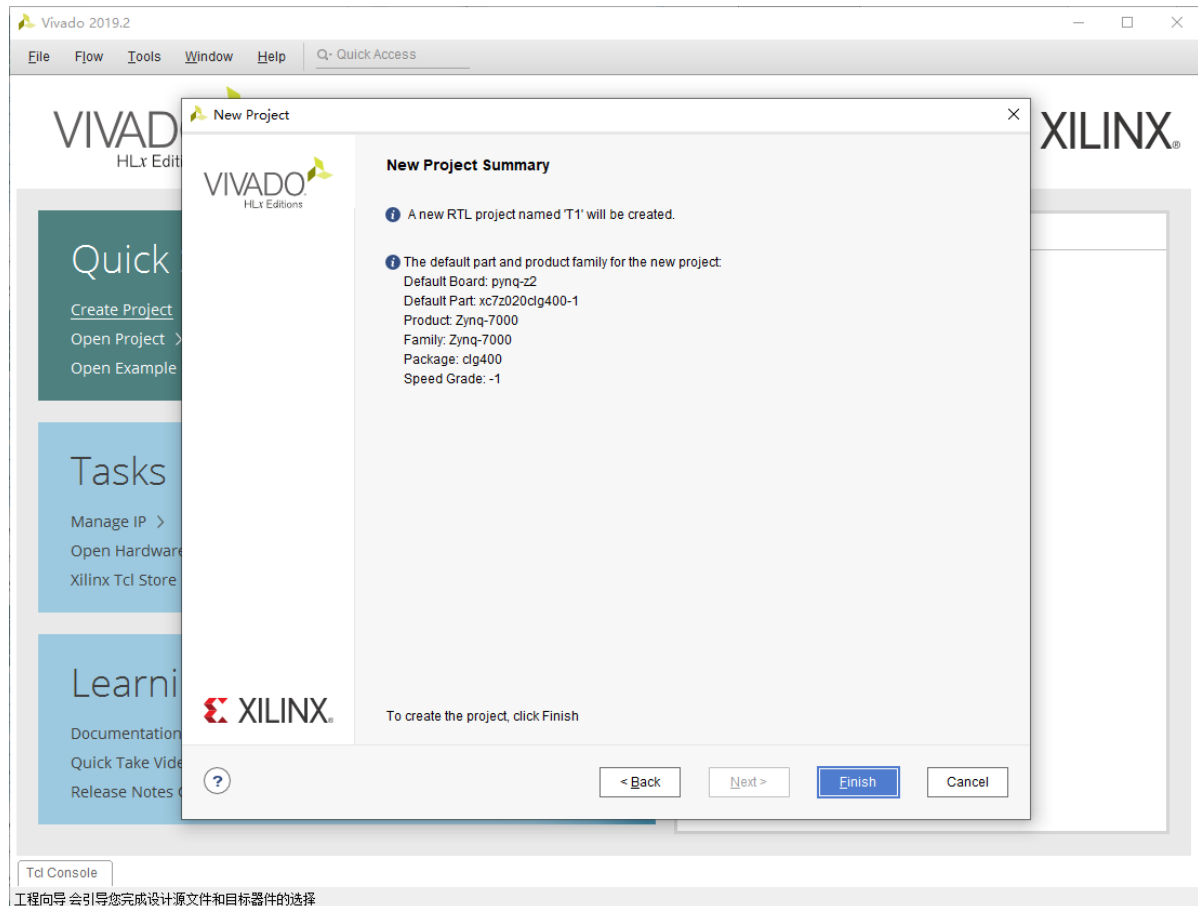
5. Select **RTL Project** option and check **Do not specify sources at this time**, and click Next.



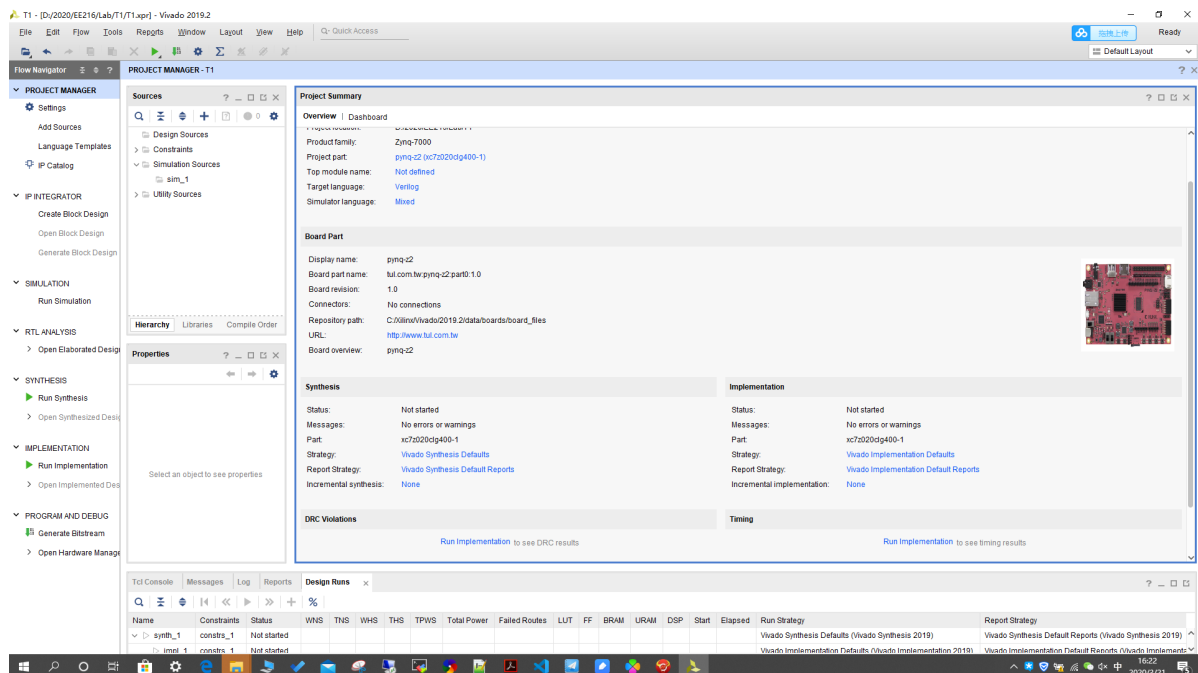
6. Select board or part (**ZCU104** for this course)



7. You will see New Project Summary dialog box. Click Finish.

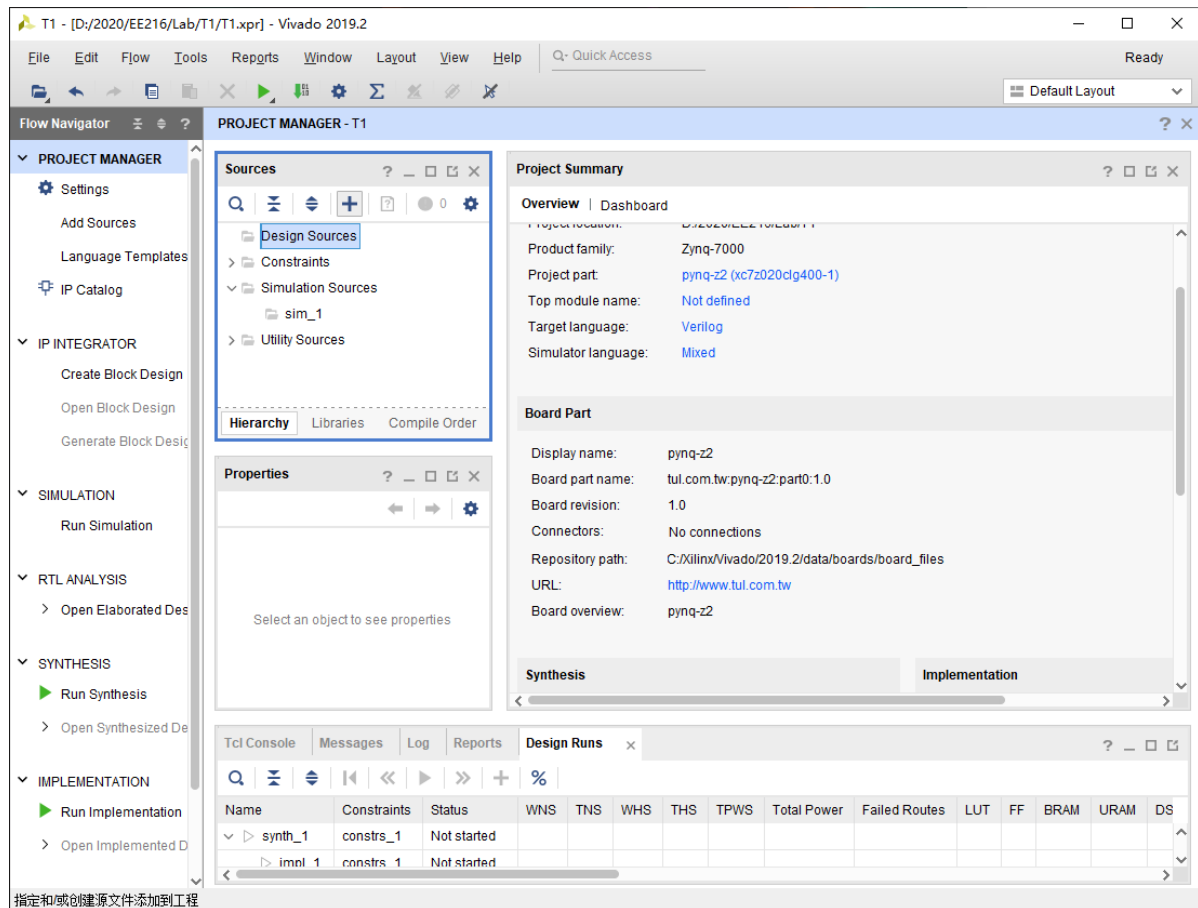


工程向导会引导您完成设计源文件和目标器件的选择

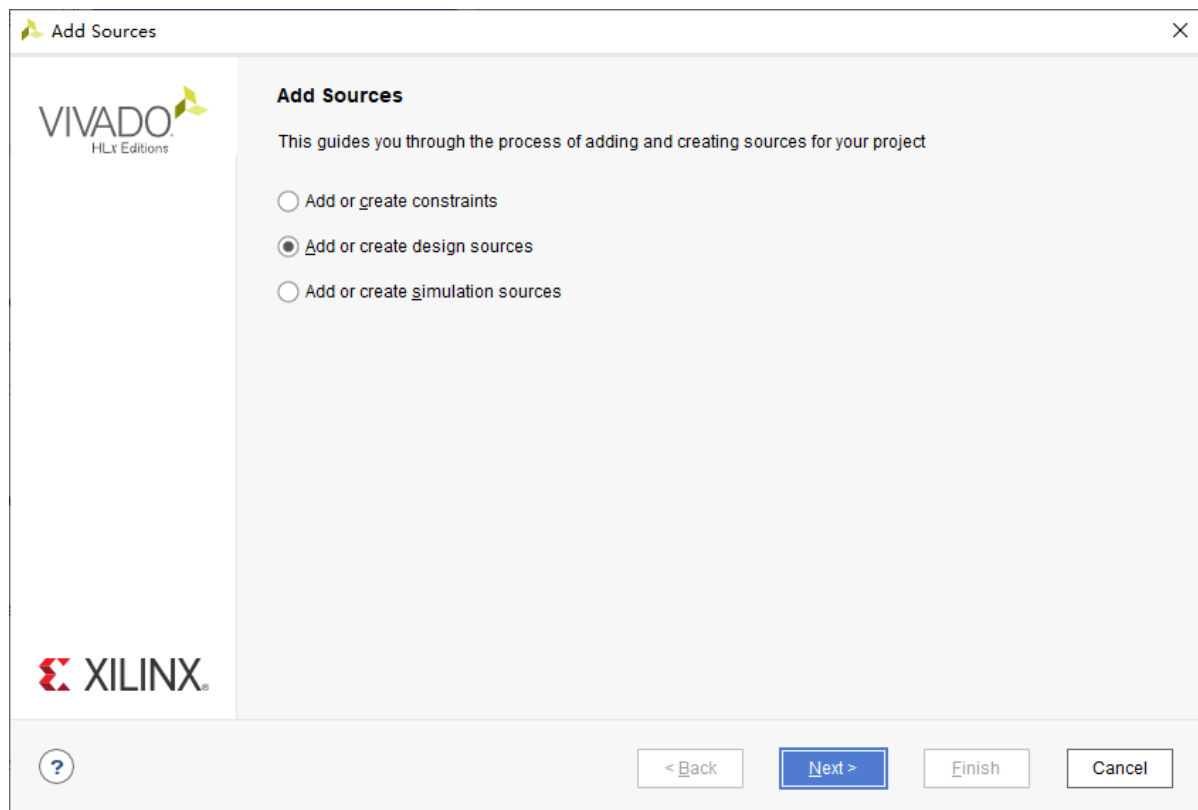


Add Source Design File

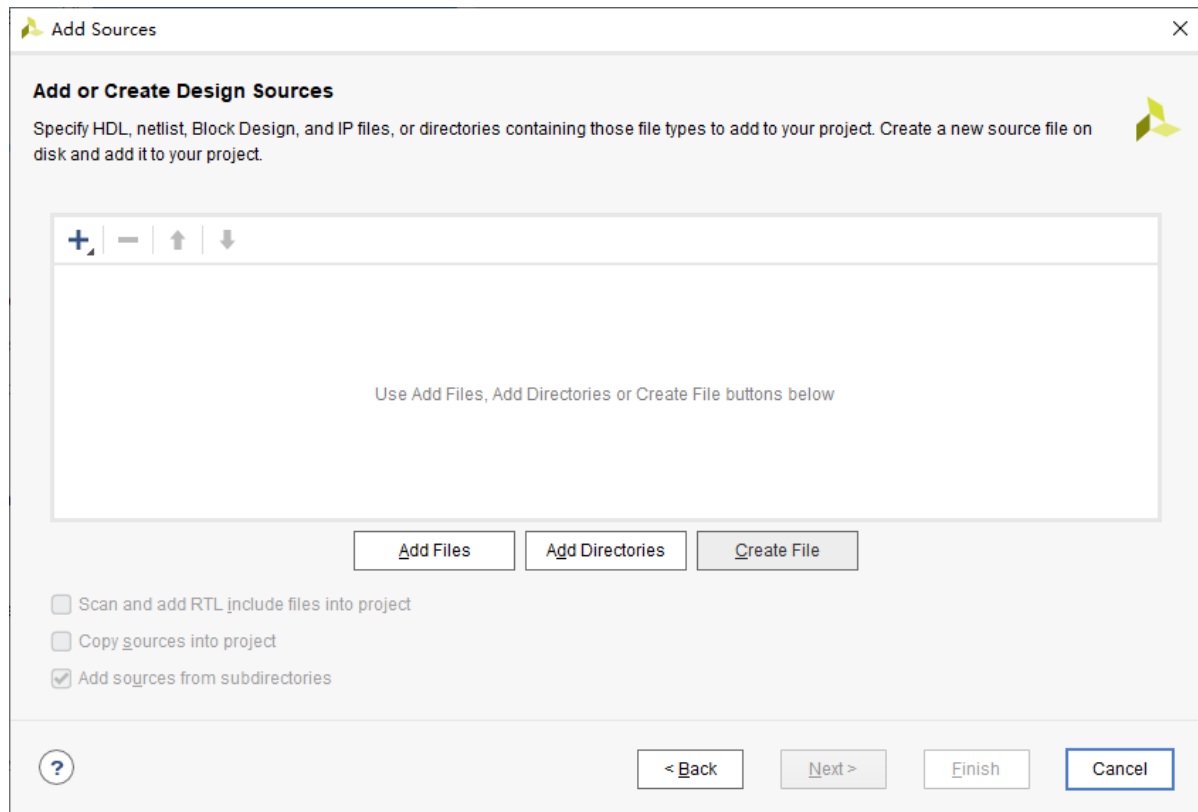
1. Click **Add Sources** under the Project Manager tasks of the Flow Navigator pane.



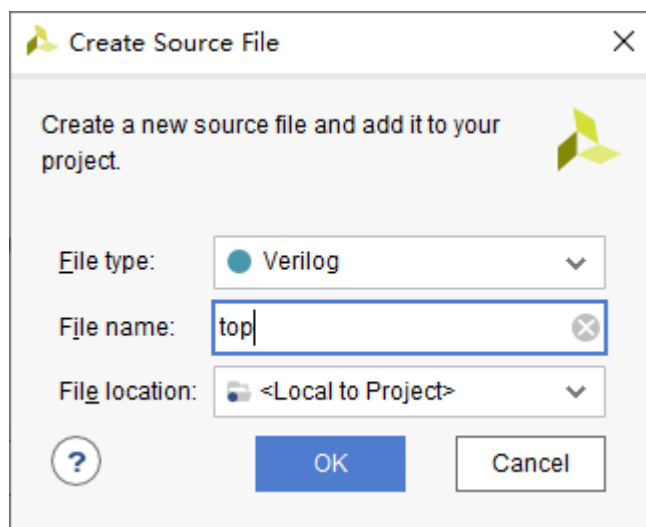
2. Select the **Add or Create Design Sources** option and click Next.



3. In the Add Sources Files form, click the **Create Files** button.



4. Set the **File type** to Verilog and input the **File name** such as top, and click OK.



5. Click **Finish**.

Add Sources

Add or Create Design Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those file types to add to your project. Create a new source file on disk and add it to your project.

	Index	Name	Library	Location
	1	top.v	xil_defaultlib	<Local to Project>

☐ Scan and add RTL include files into project
☐ Copy sources into project
☒ Add sources from subdirectories

- In I/O Port Definition of the pop-up Define Module, enter the design module required port, and set the port, if the port is bus type, check the Bus option, and set the MSB and LSB based on the bus width. Click OK when finished.

Define Module

Define a module and specify I/O Ports to add to your source file.
 For each port specified:
 MSB and LSB values will be ignored unless its Bus column is checked.
 Ports with blank names will not be written.

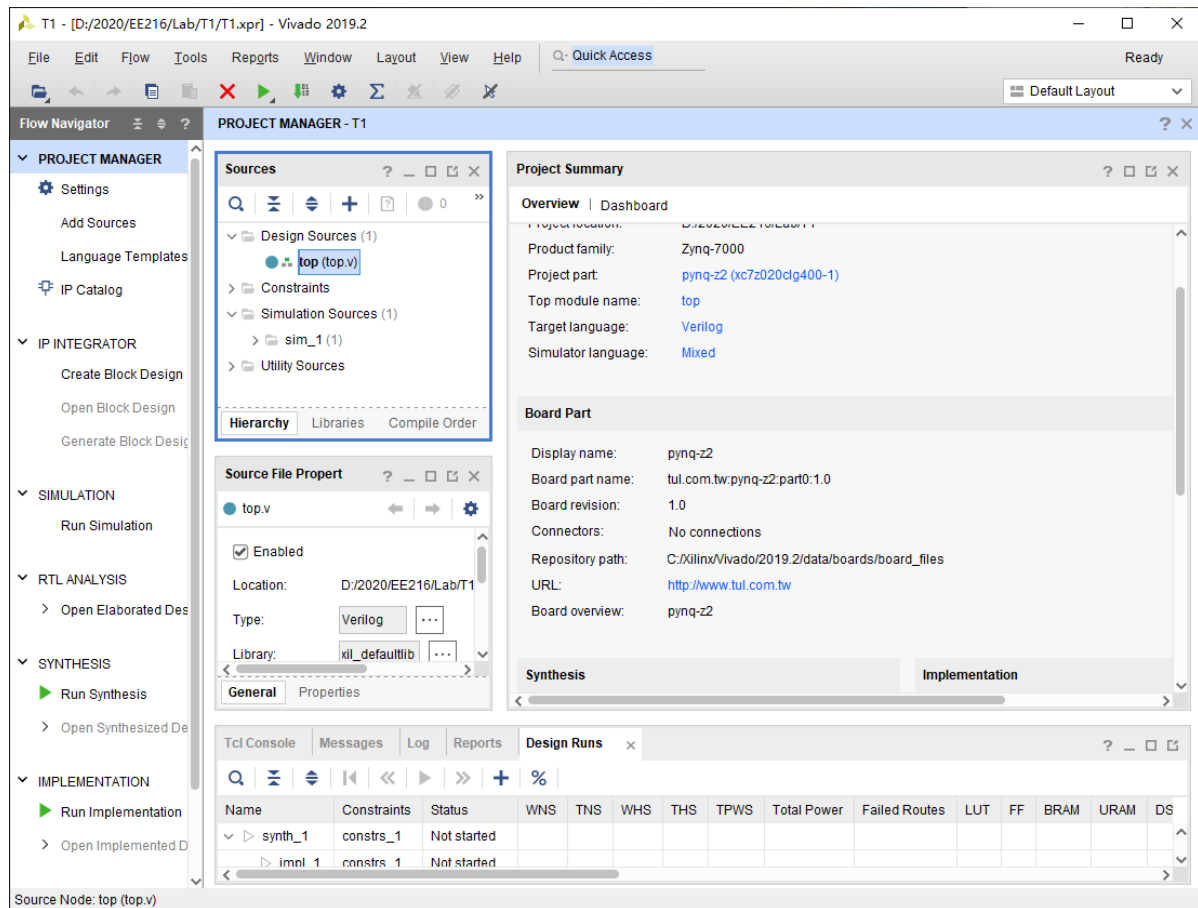
Module Definition

Module name:

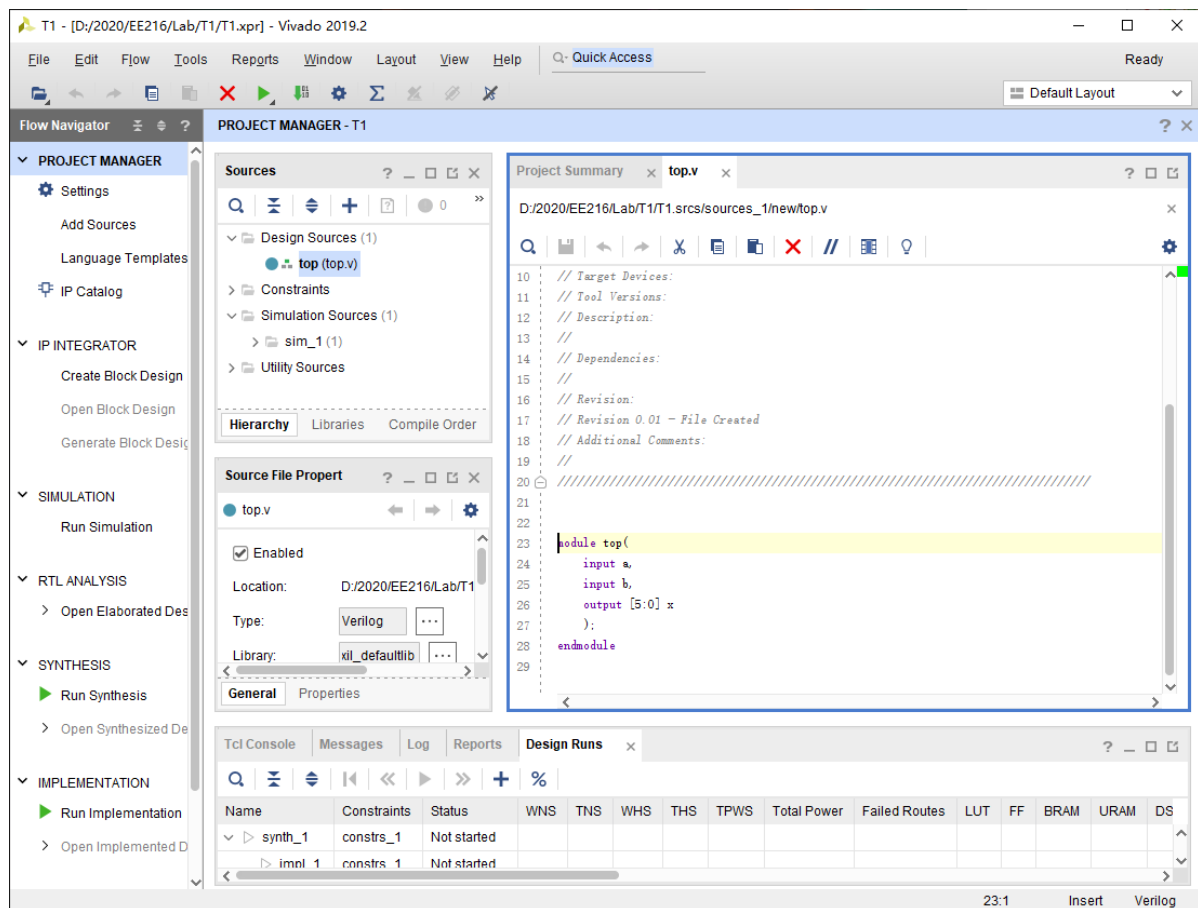
I/O Port Definitions

Port Name	Direction	Bus	MSB	LSB
a	input	☐	0	0
b	input	☐	0	0
x	output	☒	5	0

- Double click the new created source file top and open it.



8. Now you can type in your own Verilog design code.



9. Edit the file as below. Save the file

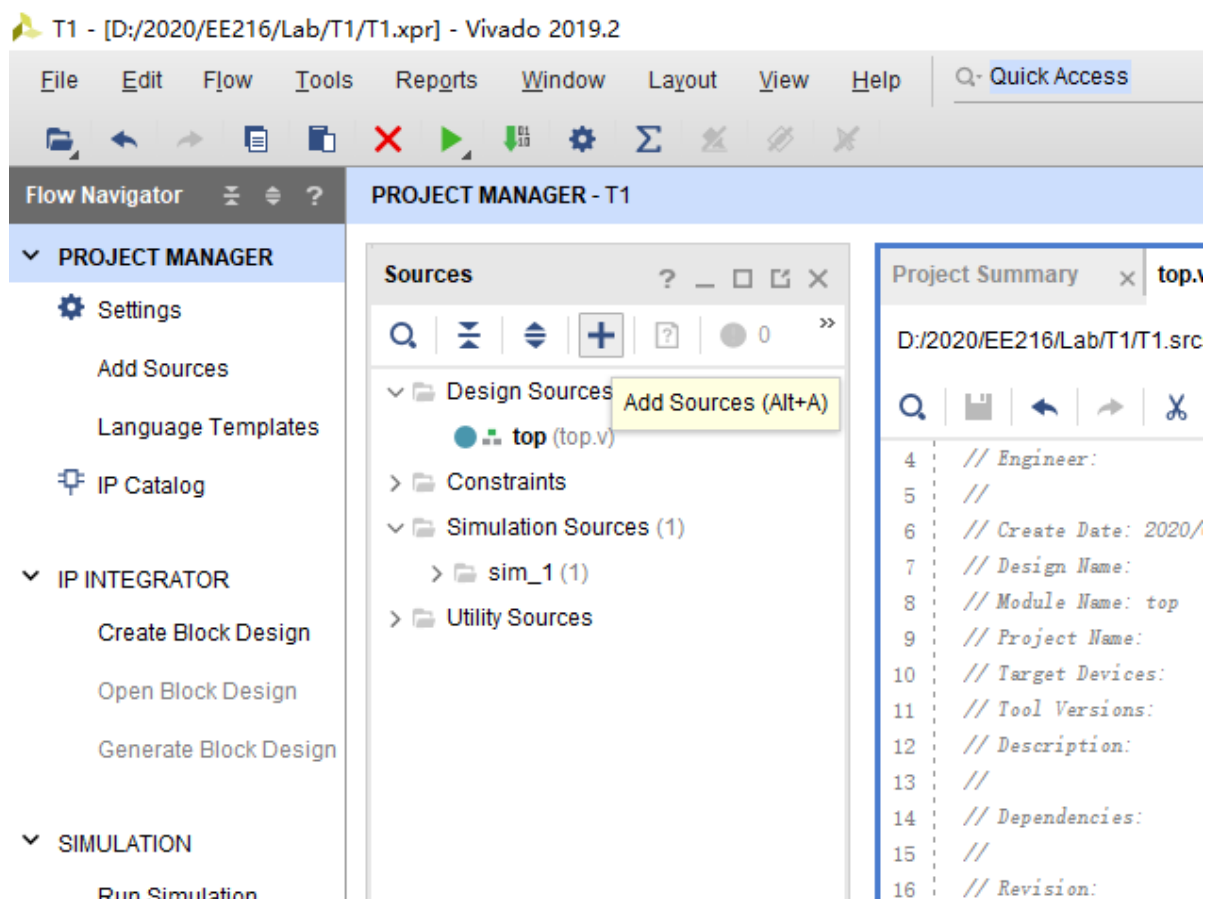
```

module top(
    input a,
    input b,
    output [5:0] x
);
assign x[0] = a | b;
assign x[1] = a | (~b);
assign x[2] = a ^ b;
assign x[3] = a ^ (~b);
assign x[4] = a & b;
assign x[5] = a & (~b);
endmodule

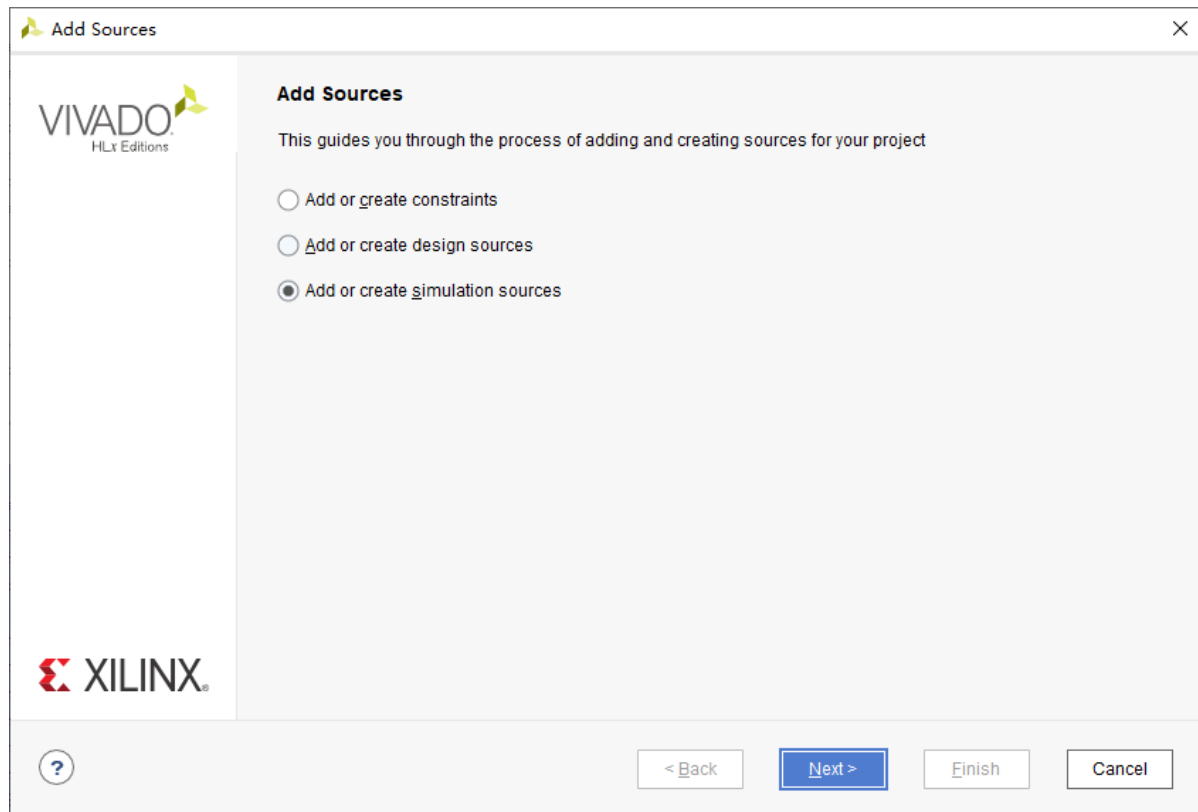
```

Behavioral Simulation

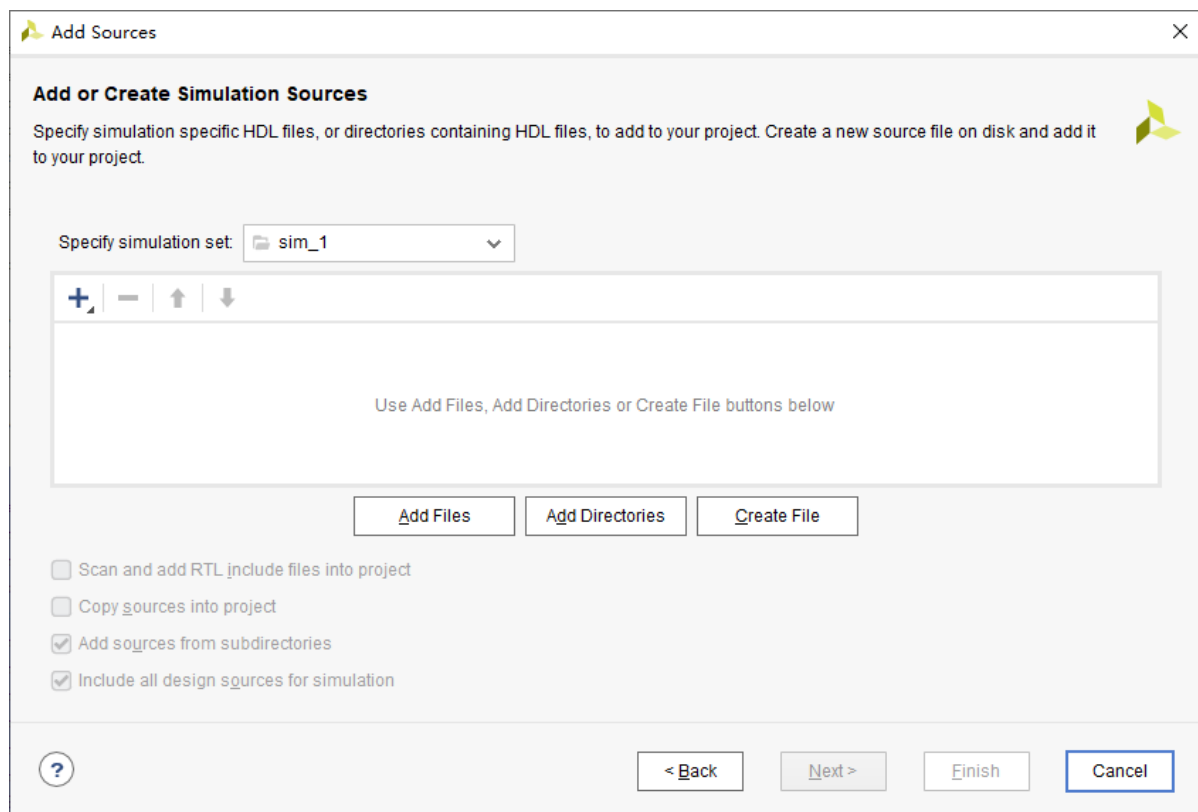
1. Click **Add Sources** under the Project Manager tasks of the Flow Navigator pane.



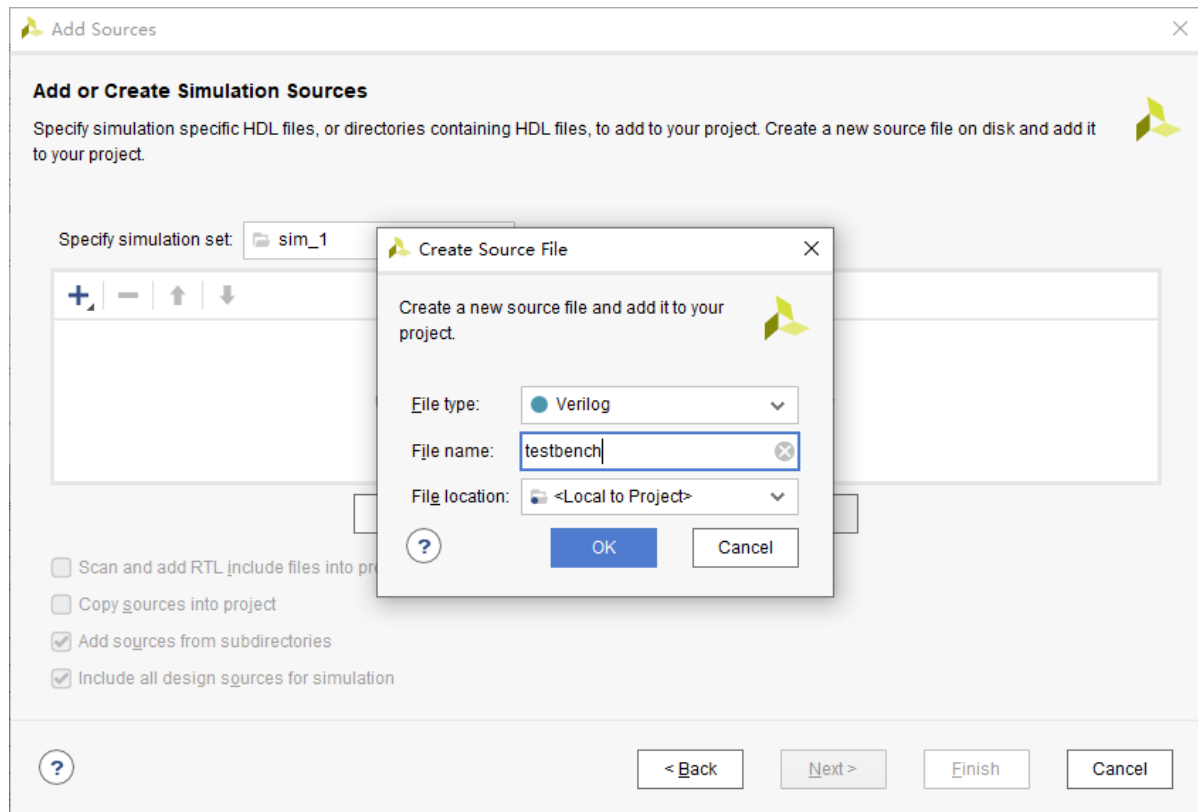
2. Select the **Add or Create Simulation Sources** option and click Next.



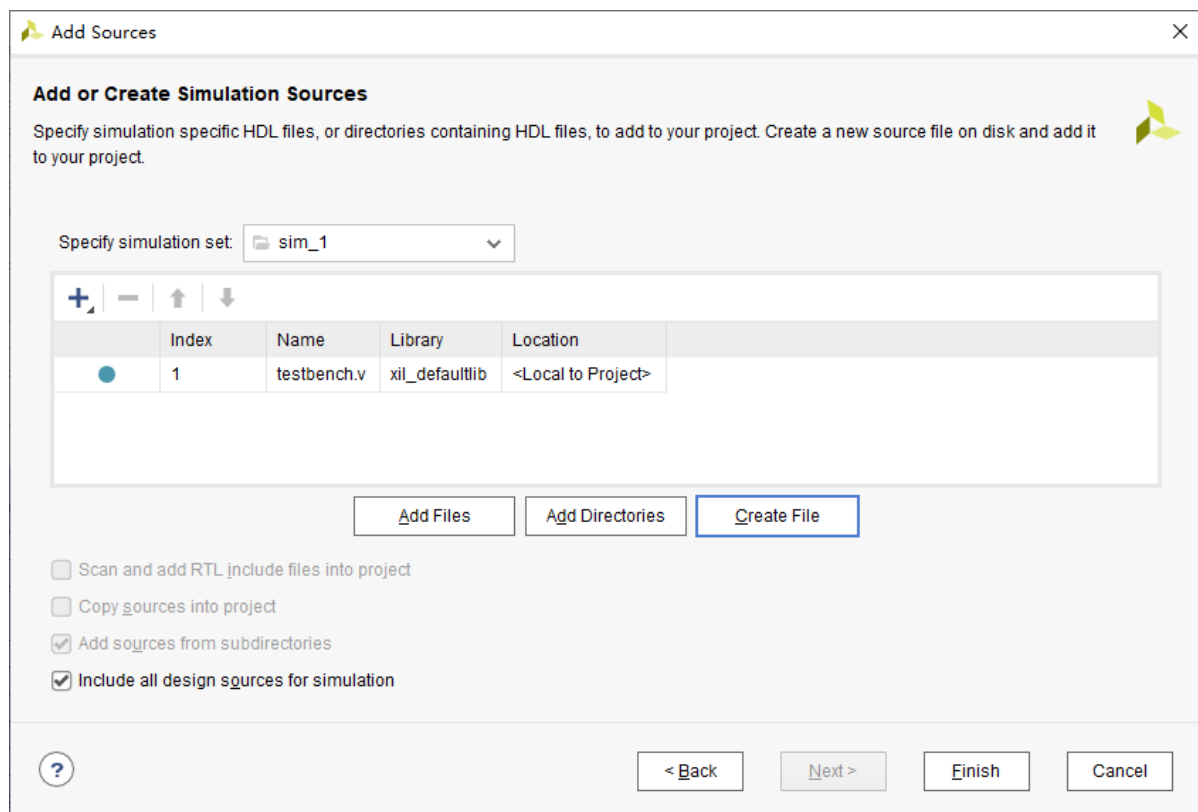
3. In the Add Sources Files form, click the **Create File** button.



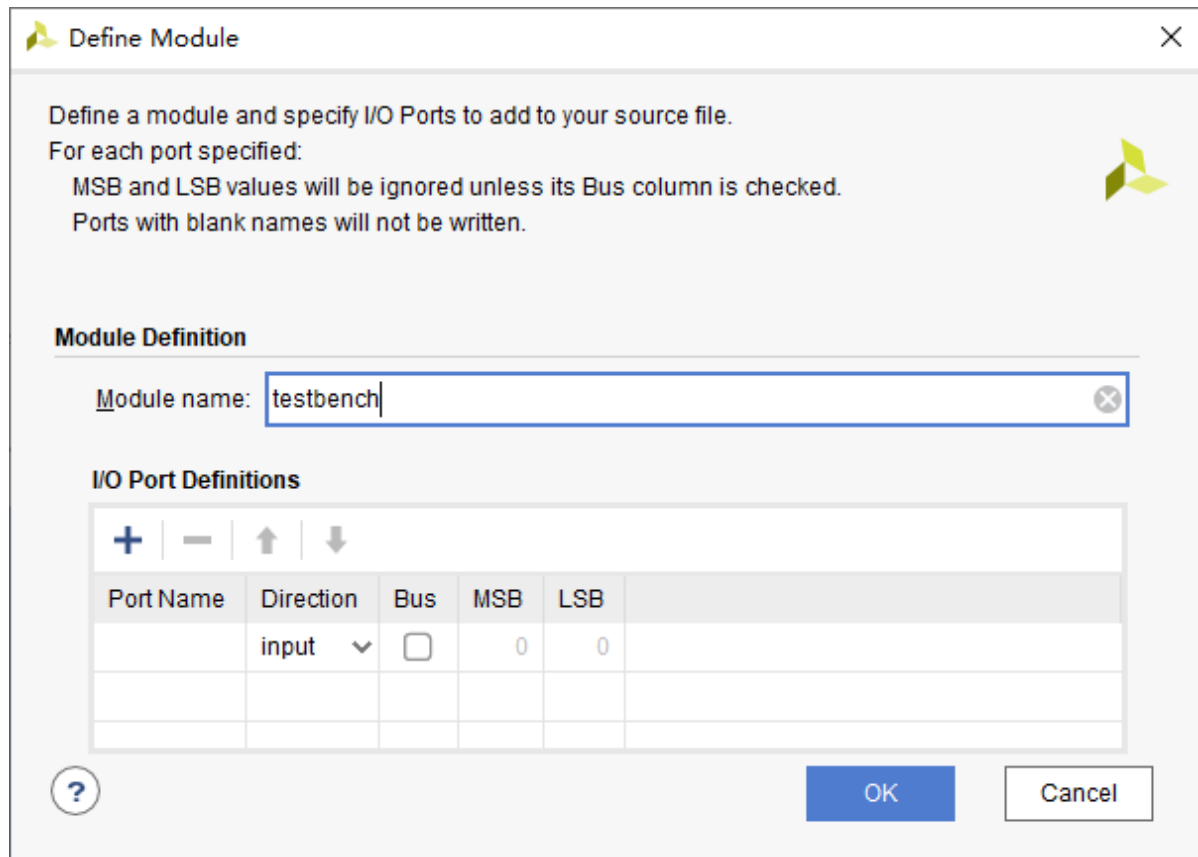
4. Set the **File type** to Verilog and input the **File name**, and click OK.



5. Click **Finish**.



6. Click **OK**.



Define a module and specify I/O Ports to add to your source file.
 For each port specified:
 MSB and LSB values will be ignored unless its Bus column is checked.
 Ports with blank names will not be written.

Module Definition

Module name:

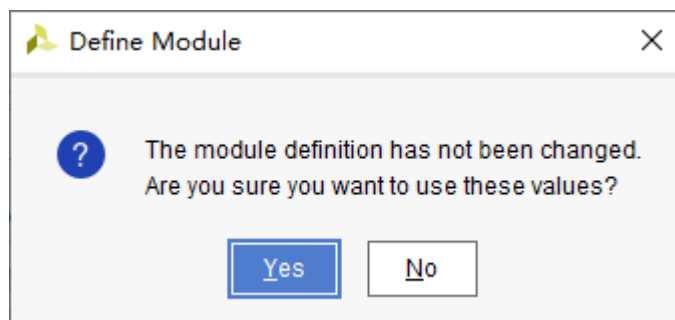
I/O Port Definitions

+
-
↑
↓

Port Name	Direction	Bus	MSB	LSB
	input	☐	0	0

?
OK
Cancel

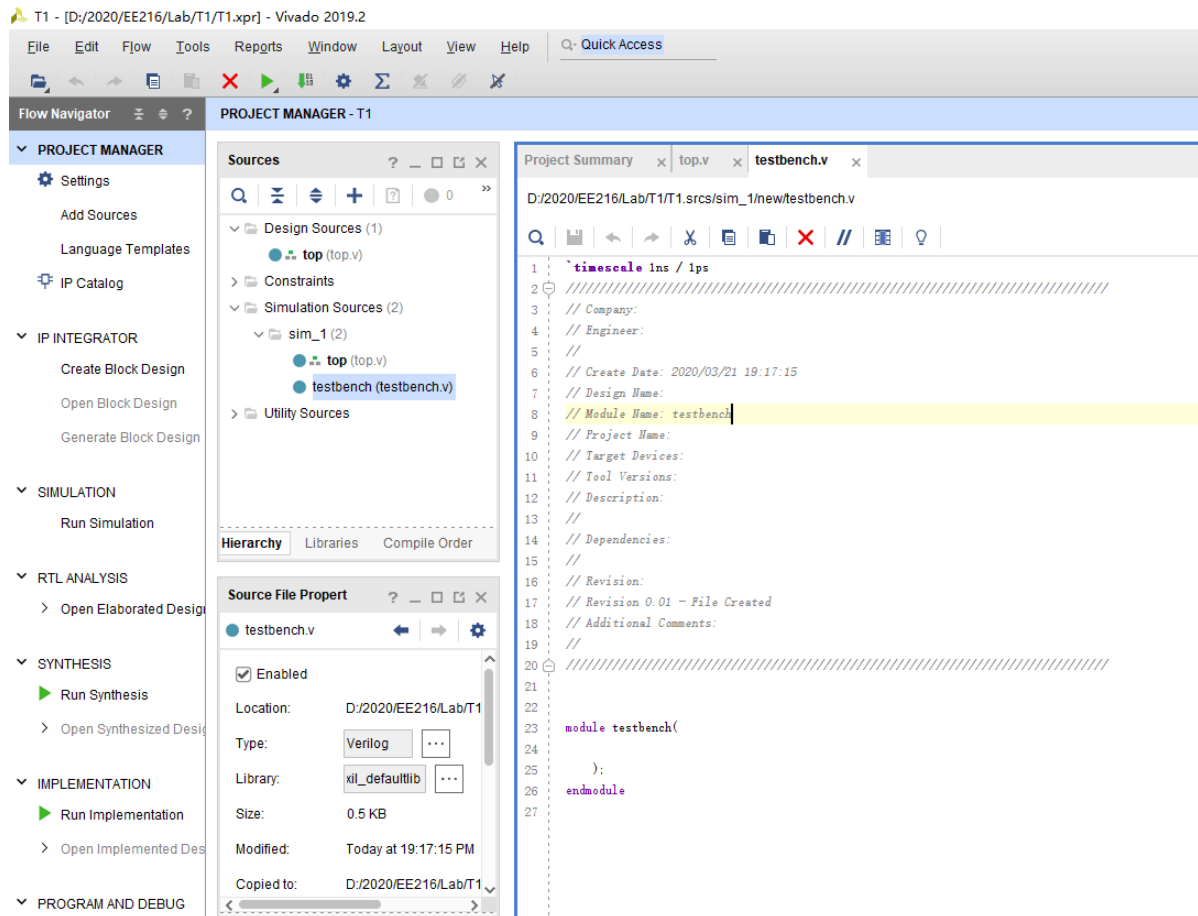
7. Click **Yes**.



The module definition has not been changed.
 Are you sure you want to use these values?

Yes
No

8. Double click the new created source file testbench and open it. Then you can type in the test code.



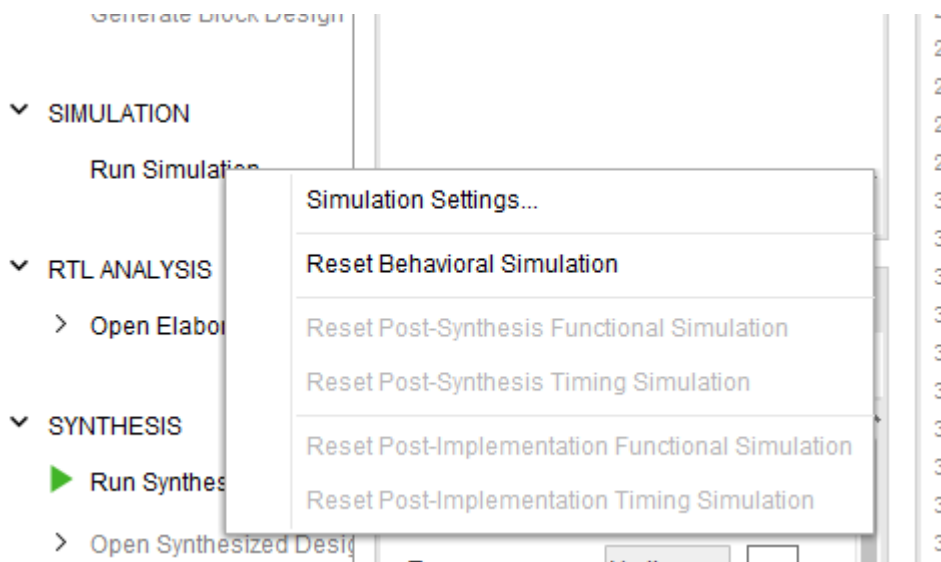
9. Edit the file as below. Save the file.

```

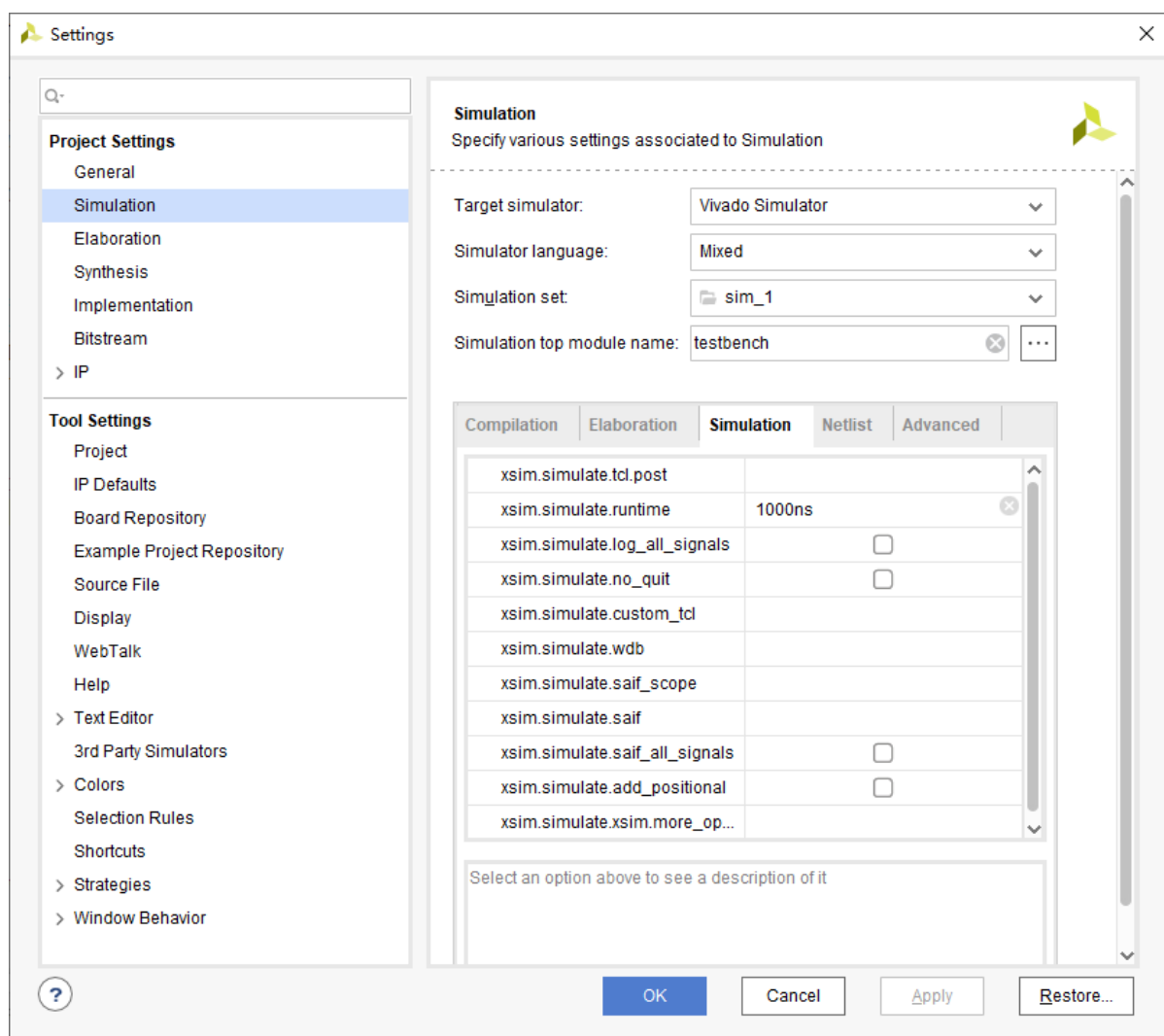
module testbench();
    reg a;
    reg b;
    wire [5:0] x;
    top uut_top(
        .a(a),
        .b(b),
        .x(x)
    );
    initial begin
        a = 0;
        b = 0;
        #10;
        a = 0;
        b = 1;
        #10;
        a = 1;
        b = 0;
        #10;
        a = 1;
        b = 1;
        #10;
    end
endmodule

```

10. Select **Simulation Settings** under the Project Manager tasks of the Flow Navigator pane. A Project Settings form will appear showing the Simulation properties form.



11. Select the **Simulation** tab, and set the **Simulation Run Time value** to 1000 ns and click OK.



12. Click on **Run Simulation > Run Behavioral Simulation** under the Project Manager tasks of the Flow Navigator pane.

Run Simulation

Run Behavioral Simulation

Run Post-Synthesis Functional Simulation

Run Post-Synthesis Timing Simulation

Run Post-Implementation Functional Simulation

Run Post-Implementation Timing Simulation

- The testbench and source files will be compiled and the Vivado simulator will run (assuming no errors). You will see a simulator output similar to the one shown below.

